

Languages have some fundamental characteristics. They are mostly contextual, and have both explicit and implicit meanings. Syntax and semantics are at their core. Also, they are usually understood along with their cultural underpinnings.

Large language models (LLM) are basically neural networks that are designed to process large amounts of natural language data. These models are trained with vast amounts of data; they help to generate responses to user queries by understanding the context and intent of the user's question and providing responses. They use advanced deep learning algorithms to accomplish this.

Image recognition uses convolutional neural networks whereas language models are based on recurrent neural networks. The feedback is used for training the models. In fact, LLMs use a combination of various neural networks to provide the most approximate output.

Factors that led to the growth of LLMs

Initially developed as rule-based systems, LLMs were first used to translate text word by word. Over time, they started using sophisticated models. For example, GPT-3 used 175 billion parameters. The various factors that led to the growth of LLMs are:

- The availability of large amounts of training data because of the internet
 - Cheaper computing power
 - Advancements in deep learning architectures and powerful computing resources (GPUs, etc)
 - Support by big corporations like Google and Microsoft
 - Wide open source community support (OpenAI, etc)
- Organisations wanting to implement LLMs can:

- Develop and host their own model with the organisational data

Responsible AI

Responsible AI is the practice of using AI for ethical purposes. The process of architecting and building AI systems should be beneficial to businesses and communities.

The key aspects of responsible AI are:

- Governance model and human oversight
- Harm-prevention first approach
- Amenable for verification and validation
- Prioritising privacy

- Leverage open source transformers that are already being developed by the large open source community

How LLMs produce such good results

Recent advancements in LLMs have leveraged various advanced techniques like:

- *Positional encoding*, where words in the input can be fed into the neural network in a non-sequential way
- *Self-attention* for assigning different weights to different words in the sentence. The weight signifies the importance of a word in that context in relation to the other words
- *In-context learning (ICL)* for understanding the context of the input

Use cases for LLMs

LLMs have a plethora of use cases in various industries.

- Customer service
 - Chatbots and virtual assistants
 - Question and answer based responses
- Code generation and debugging
 - Create code
 - Write unit test cases
 - Debug code
- Text classification
 - Sentiment analysis
- Language translation
 - Translate from one language to another

- Summarisation
 - Understanding the overall meaning and providing a succinct overview
- Content generation
 - Writing poetry
 - Marketing memos and emails
- Personalised recommendations
- Data analysis
 - Customer sentiment
 - Market analysis

Issues and limits of LLMs

Even though vast advancements have been made, LLMs have a number of limitations.

- LLMs are computer models and can only produce approximations.
- They are still not trained with the complete available data.
- They consume lots of natural resources (water) for cooling purposes, etc.
- They have security and privacy issues.
- Large models require significant costs for development and maintenance.

There is no one-size-fits-all for LLMs. Organisations need to have a proper vision and business approach. Their digital transformation agenda should be supported by business users. Data is the Holy Grail, and LLMs need to be fed with data that is as relevant as possible. That's how users can get huge benefits. **END** 🐧

 By: Phani Kiran

The author has nearly 20 years of experience in the IT industry and has worked on digital transformation initiatives.